

MAT 310, Handout 11: MGFs and Tchebysheff's Theorem (3.9, 3.11)

Moments

- k^{th} moment: $\mu'_k = E(Y^k)$
- k^{th} central moment: $\mu_k = E[(Y - \mu)^k]$

Theorem: If RVs X and Y have identical moments, i.e., $E(X^k) = E(Y^k)$ for $k \geq 1$, then X and Y have the same _____.

Question: How to compute the moments?

1. Directly: $E(Y^k) =$ _____ (for discrete RVs)
2. via moment-generating functions

Moment-Generating Function (mgf): The mgf of RV Y is given by

$$m(t) =$$

which exists if there is some $b > 0$ such that $m(t)$ is finite for $|t| \leq b$.

Why is $m(t)$ called the moment-generating function?

Recall the power series expansion: $e^x =$

Therefore, $e^{tY} =$

And so, $m(t) = E(e^{tY}) =$

Using mgfs to find moments:

Example 1. Find the mgf of $Y \sim \text{Poisson}(\lambda)$.

Example 2. For $Y \sim \text{Poisson}(\lambda)$, find $V(Y)$ using the mgf of Y .

Remark: If the mgf, $m(t)$, exists for a probability distribution, it is unique! Thus, if the mgfs for 2 RVs X and Y are equal, X and Y must have the same probability distribution.

- It follows that, if the mgf of a RV Y is the mgf for a specific distribution, then Y must have that distribution.

Recall the Binomial Theorem: $(a + b)^n = \sum_{y=0}^n \binom{n}{y} a^y b^{n-y}.$

Tchebysheff's Theorem: Let Y be a RV with mean μ and variance σ^2 . Then, for any constant $k > 0$,

$$P(|Y - \mu| < k\sigma) \geq$$

Equivalently,

$$P(|Y - \mu| \geq k\sigma) \leq$$

Remark: These bounds for the probability that any RV is within (or alternatively, outside) k standard deviations of its mean, μ , apply for every distribution.

Example 3. Suppose the mean IQ of state employees is 108 with a standard deviation of 6. Find bounds for the probability that a randomly selected employee has an IQ larger than 96 but smaller than 120.